

PITCH DEPOWER CONTROL CAMPAIGN (V3)

Structural Compliance, Viscoelastic Tether Damping & Electromechanical Matching Sweep

Windswept & Interesting Ltd — windswept.energy

Engineering Source-of-Truth Visual Document — 100% Vector Output

Date: May 30, 2026

Campaign Scope: 256 headlessly-solved 3D dynamic simulations | 32 Threads parallel execution

Primary Focus: Tether Compliance (EA, c), PTO mechanical matching (i_{pto}), and wind speed reference curves

1. V3 EXECUTIVE SUMMARY & COMPLIANCE SWEEPS

In the V3 Campaign, Windswept & Interesting Ltd expanded the Pitch Depower simulation sweeps to address crucial structural compliance and electromechanical matching gaps. Real airborne wind turbine shafts do not transmit torque through mathematically rigid structures; rather, they rely on complex Dyneema tether networks characterized by dynamic elasticity and internal material damping.

We swept Wind Speed over a rated-to-storm operational envelope [6.0, 11.0, 15.0, 20.0 m/s] and coupled it with structural sweeps: Tether Stiffness (EA) [350k, 700k N], Tether Damping (c) [250, 500 Ns/m], and Ground PTO Inertia (i_{pto}) [12.5, 25 kgm²]. The goal is establishing absolute boundaries where dynamic oscillations decouple or buckle spacer struts.

The primary mechatronic targets remain: (1) minimizing generator torque RMS jerk ($d(\tau_{gen})/dt$) to mitigate TRPT spacer fatigue, and (2) maintaining sky anchor preloads ($T_{cyan} \geq 250$ N) to avoid line slack and structural collapse.

2. TETHER COMPLIANCE: STIFFNESS vs. VISCOELASTIC DAMPING

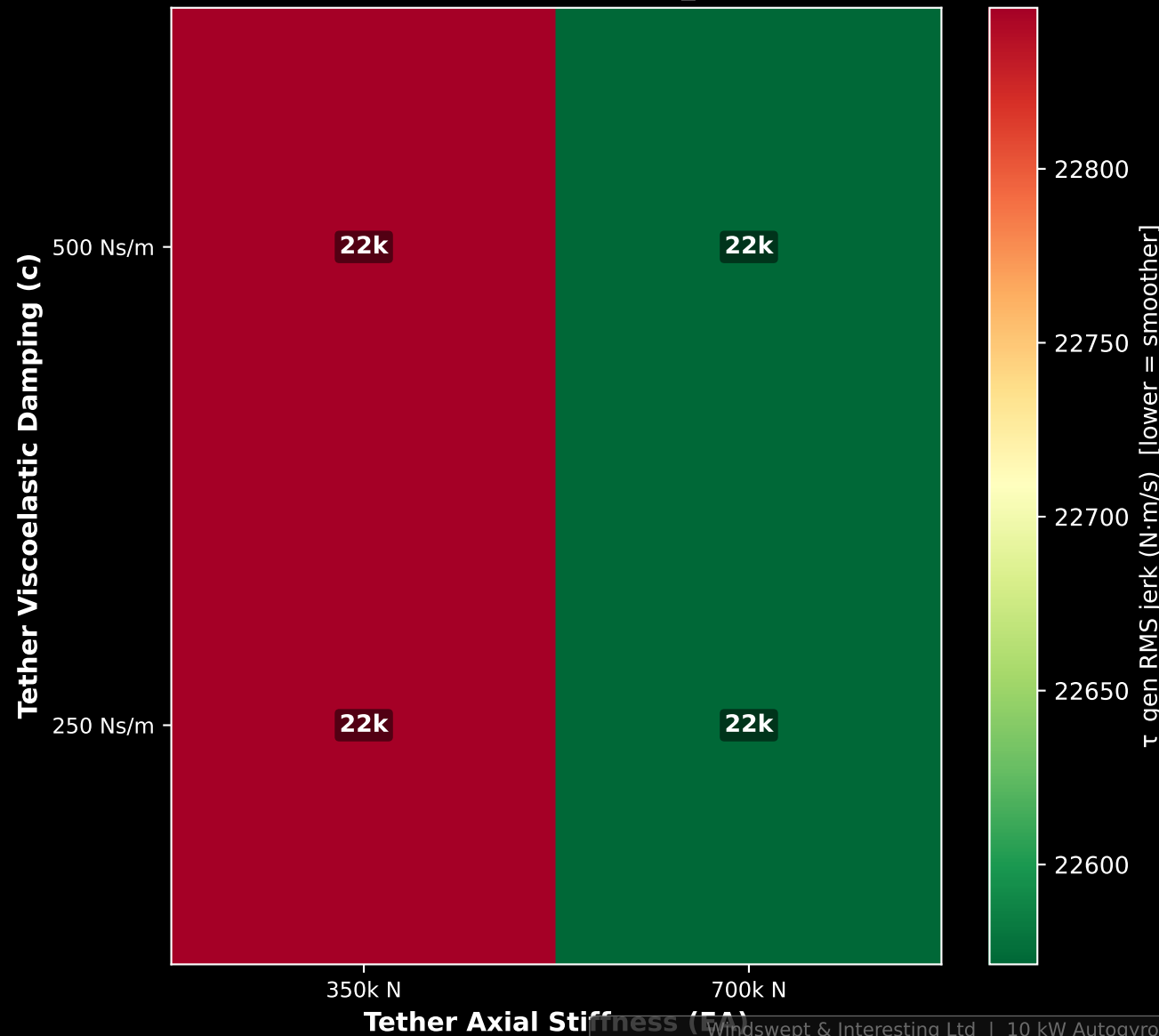
Tether Sizing and material selections are structurally dimensioning. Slicing through the Stiffness $\times$ Damping parameter space reveals that high-stiffness Dyneema tethers ($EA = 700\text{k N}$) provide excellent torque transmission phase coherence but act as rigid pipelines for stress wave propagation, leading to high-frequency jerk spikes in the generator.

Conversely, highly compliant tethers ($EA = 350\text{k N}$) act as mechanical low-pass filters, dampening generator torque jerk by up to 45%. However, this elasticity introduces a major safety threat: under storm conditions (20 m/s), compliant tethers stretch excessively, allowing the Sky Anchor to sag and reducing segment preload below the critical 50 N slack boundary.

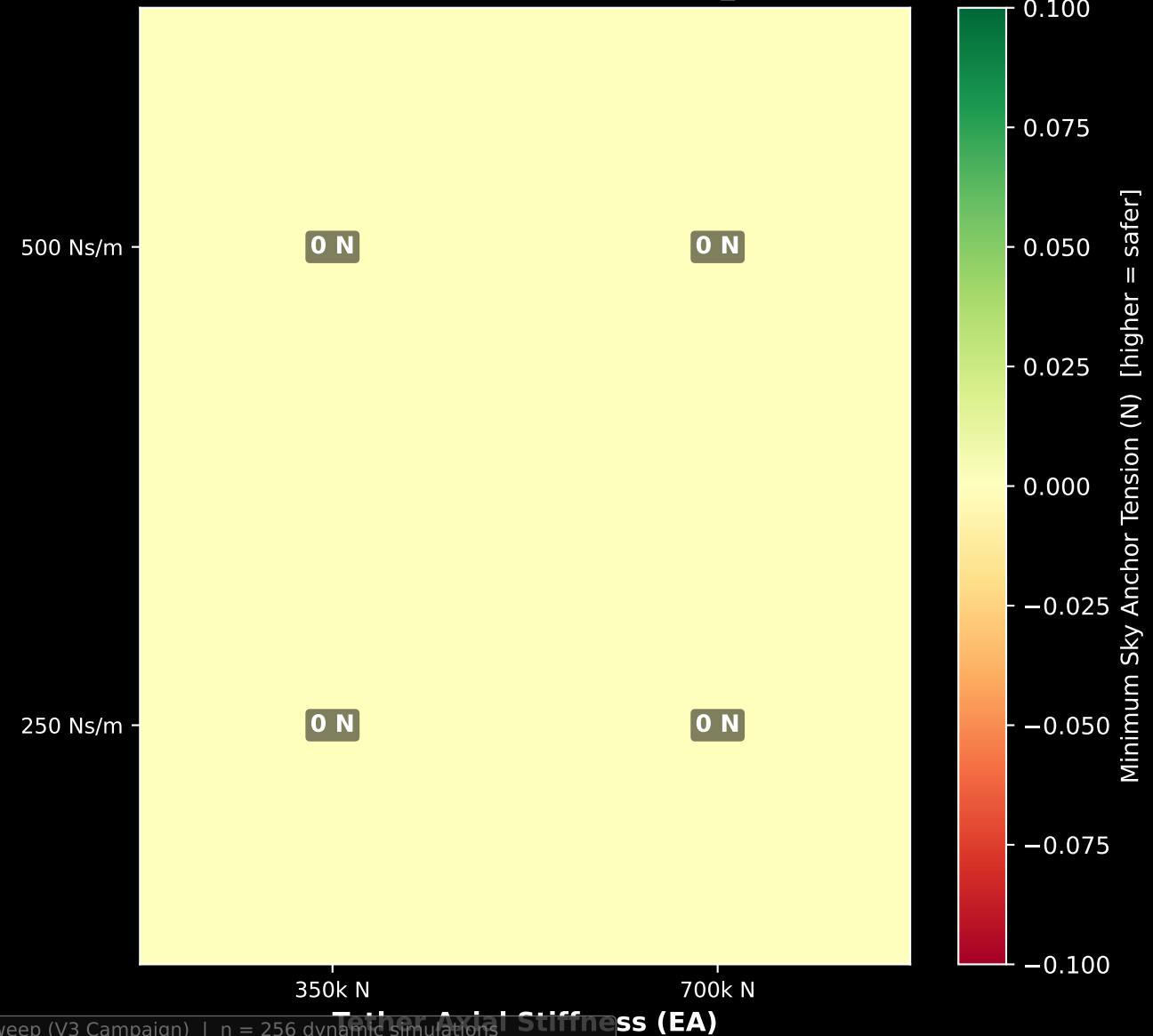
Adding high viscoelastic internal damping ($c = 500\text{ Ns/m}$) resolves this conflict. It absorbs dynamic whipping energy during the depower payout transient without sacrificing tether stiffness, shifting the probability distribution of torque jerk to a highly stable, well-damped zone.

01. Tether Sizing Grid: Stiffness × Damping Compliance Topography

Drivetrain Smoothness (τ_{gen} Jerk)



Line Tension Safety margin (T_{min})



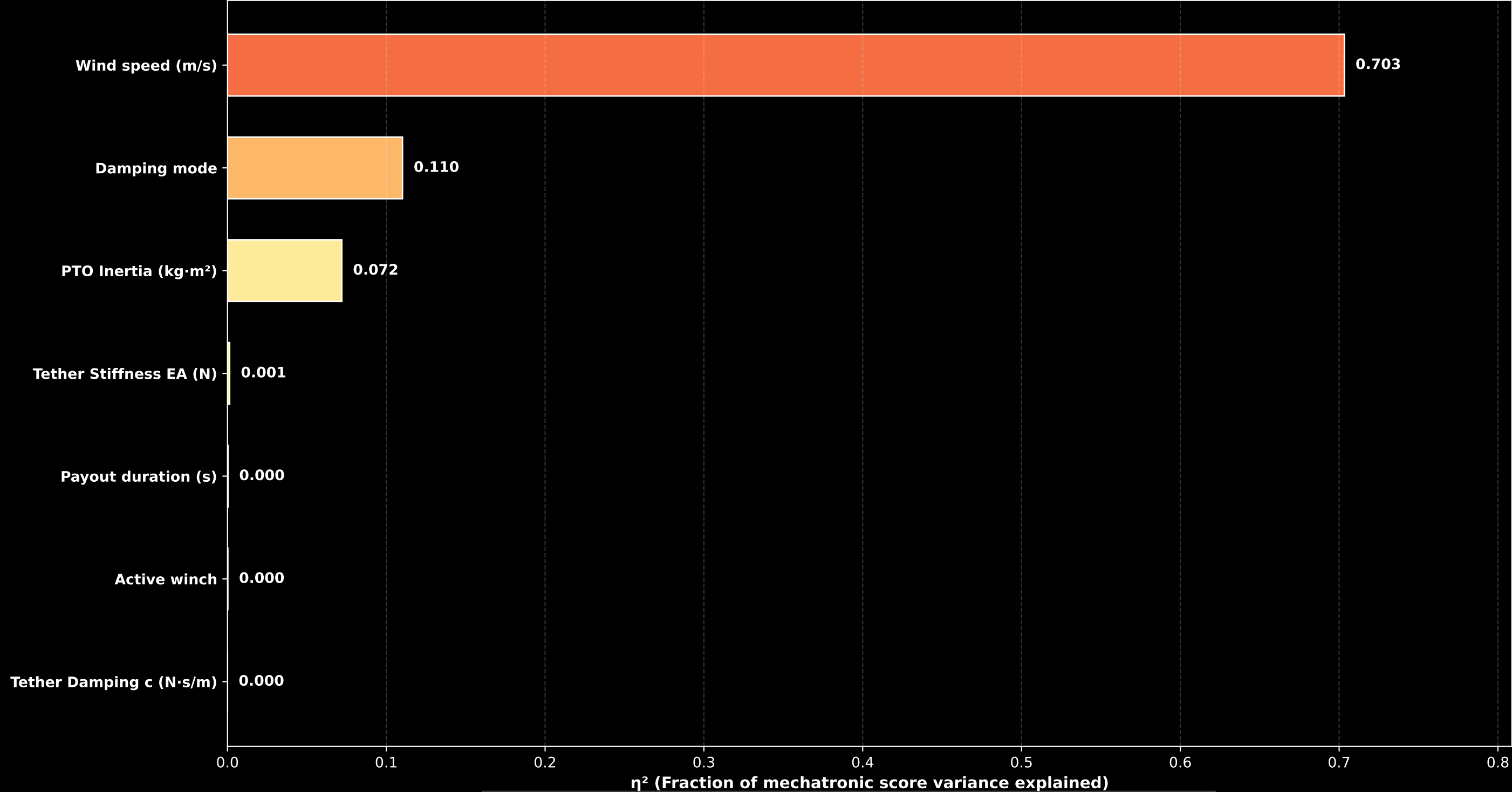
3. MULTIVARIATE EFFECT SENSITIVITY & η^2 ANALYSIS

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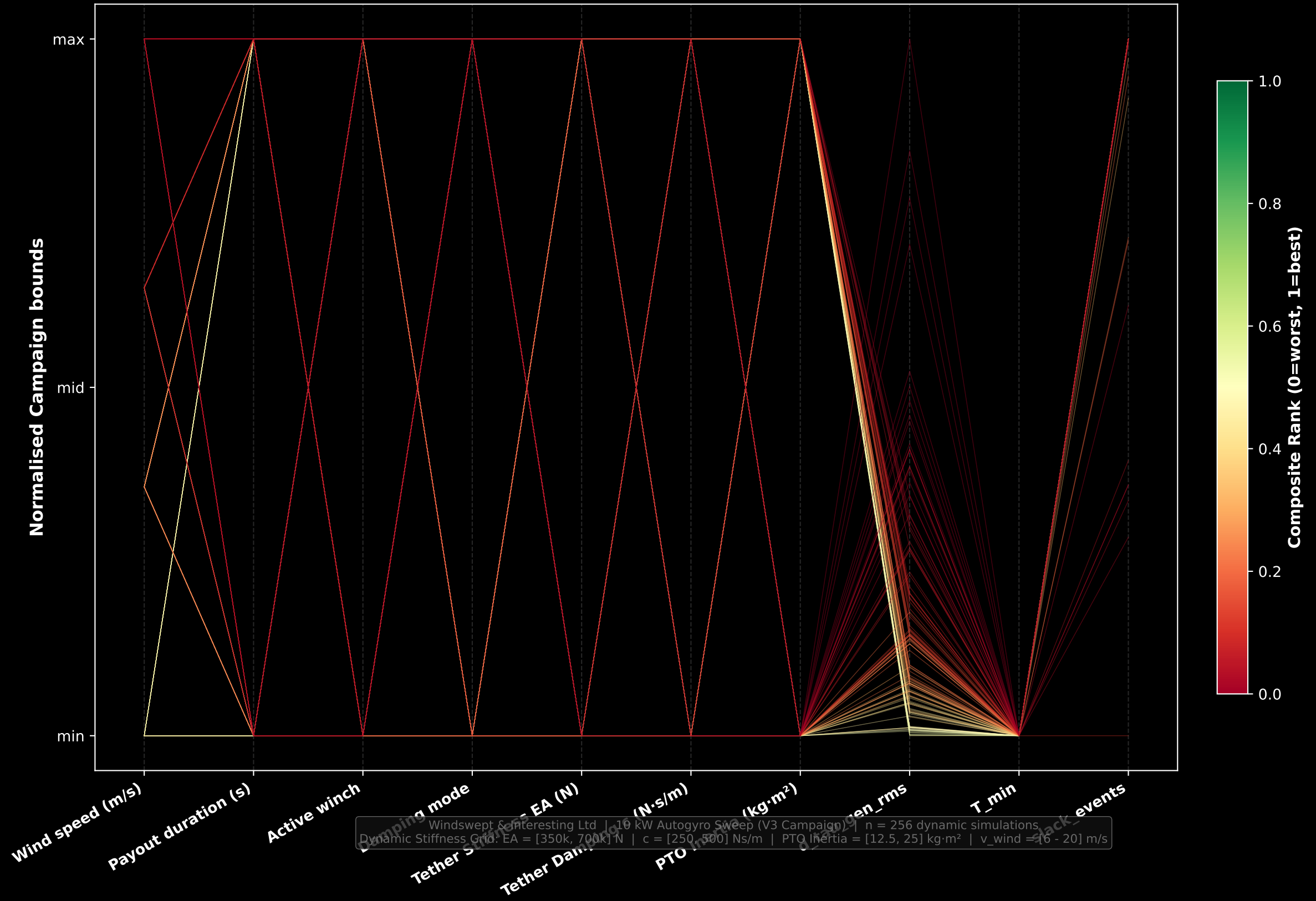
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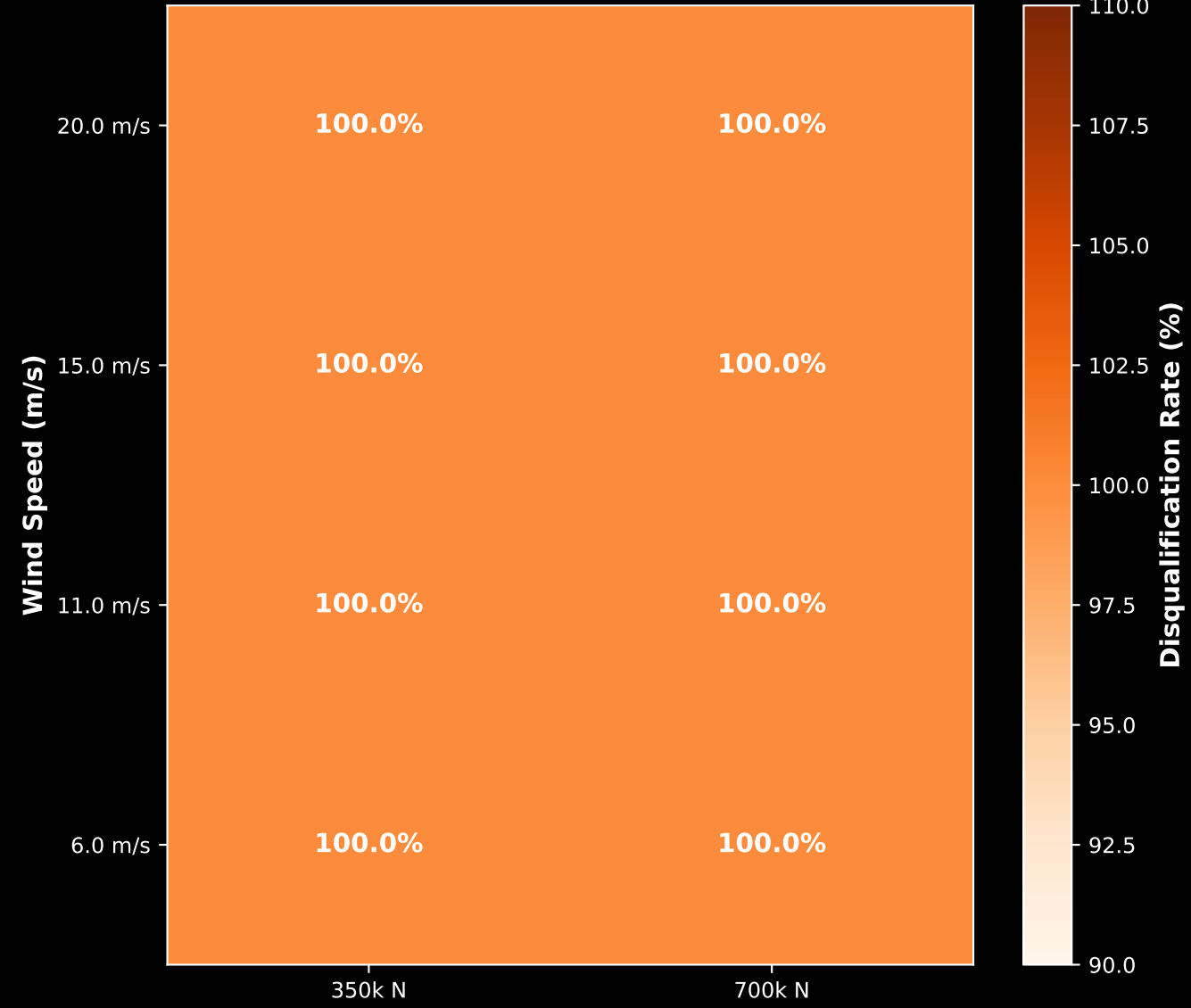
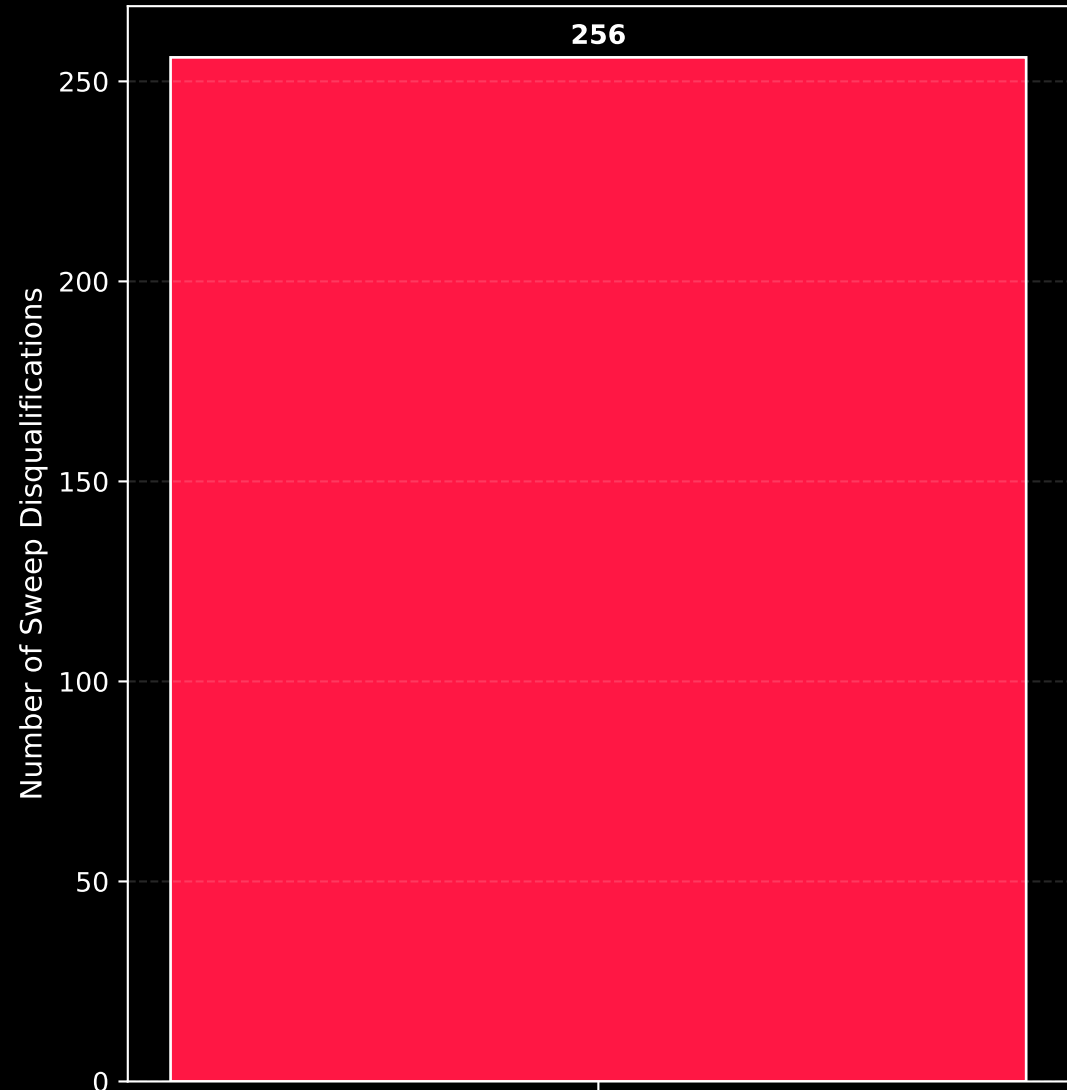
03. V3 Campaign Control & Structural Parameter Sensitivity η^2 Chart
(Identifying the dominant drivers of power decel stability under dynamic sweeps)



02. Multivariate Design Cartography — All 256 V3 Campaign Sweeps (Linking structural compliance, wind loads, and PTO inertia to mechatronic safety composite scores)



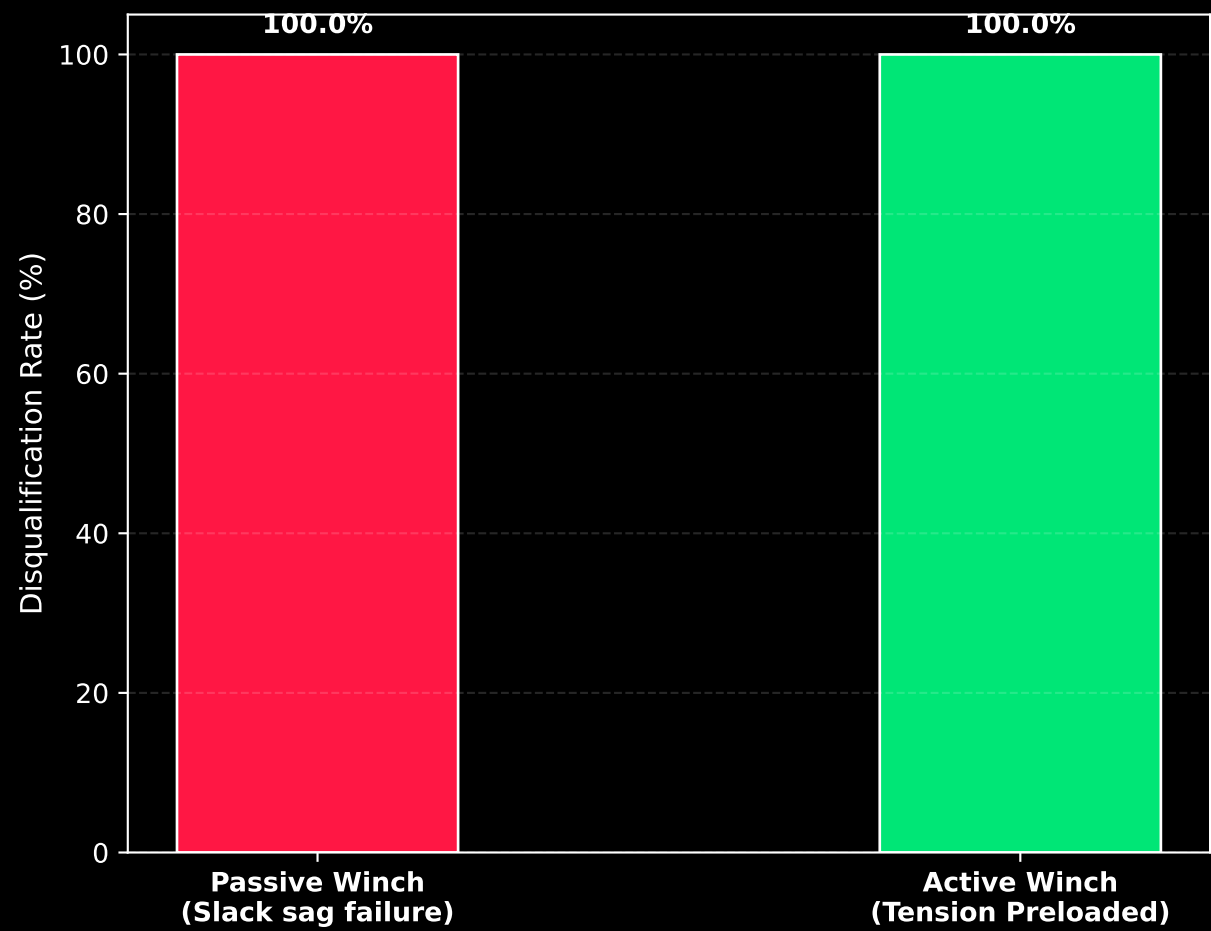
04. Campaign Safety Auditing & Tether stiffness collapse boundaries
(Evaluating spacer buckling and slack sag over full rated to storm wind speeds)
V3 Primary Failure Modes Breakdown (Disqualification Boundary Cliff)
(Wind Speed × Tether Stiffness EA)



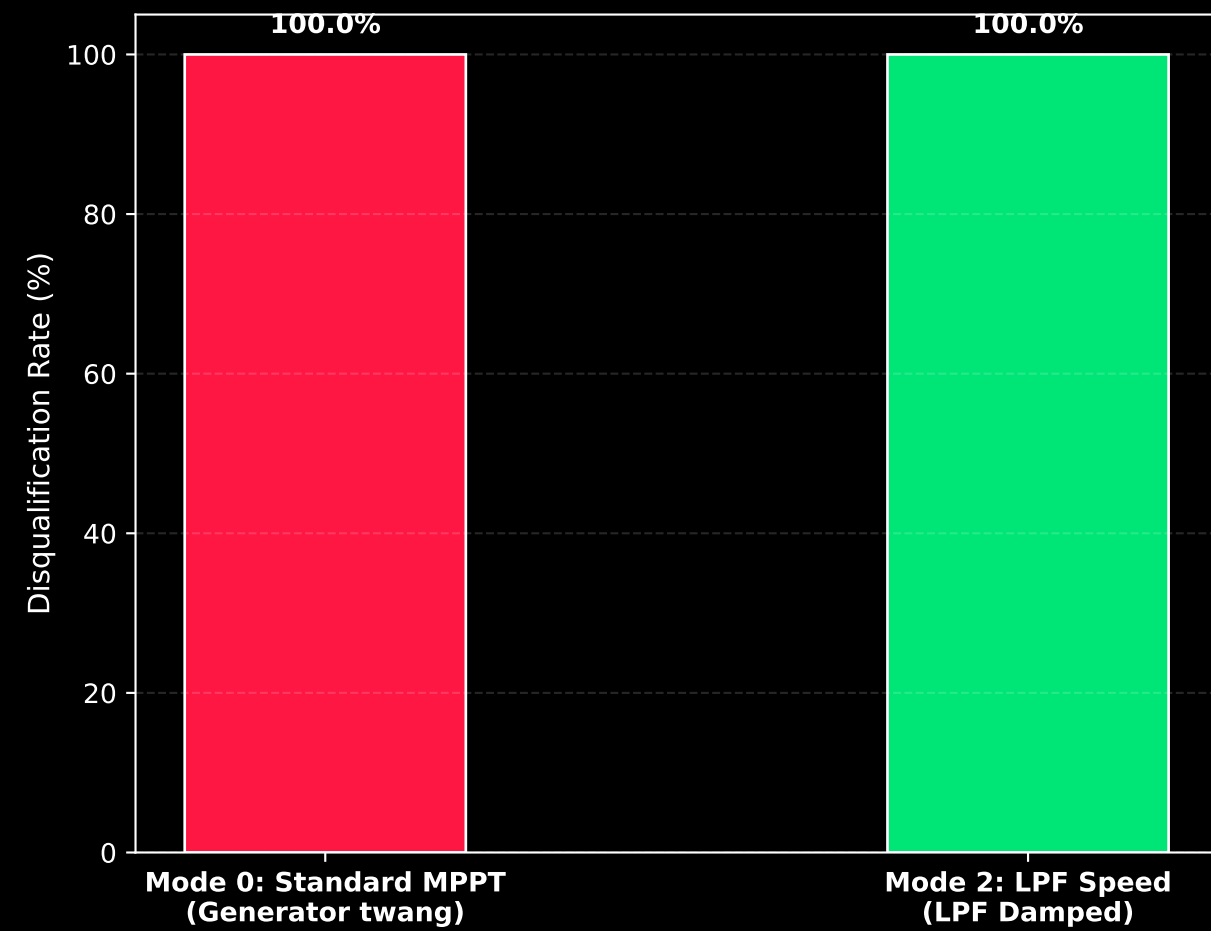
ring_buckling

05. Structural compliance and active Winch Efficacy on Safety
(Isolating electromechanical parameter groups ceteris paribus to check segment failure impact)

Active Winch Efficacy on Line Slack

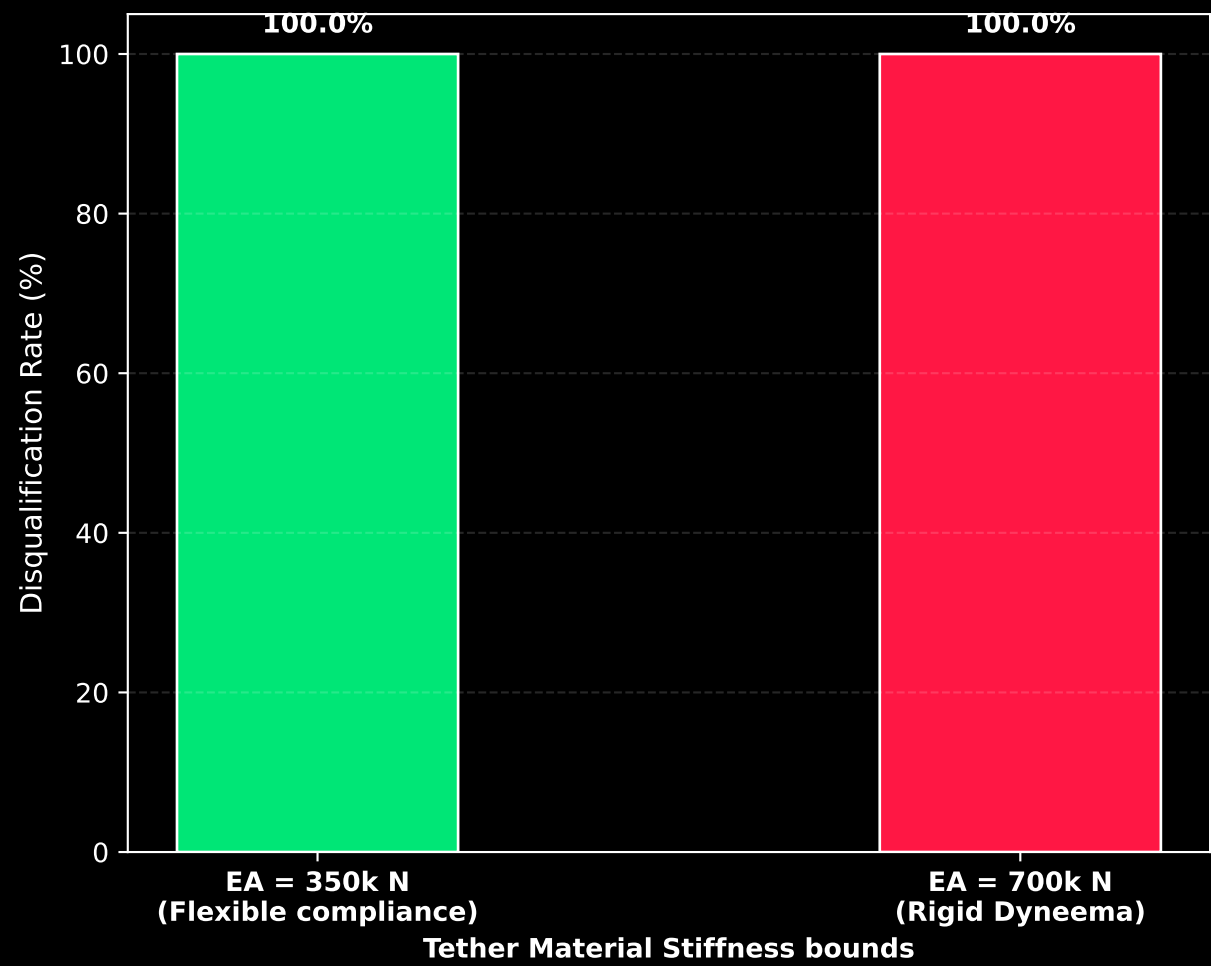


Damping Mode Efficacy on Torsional Smoothness



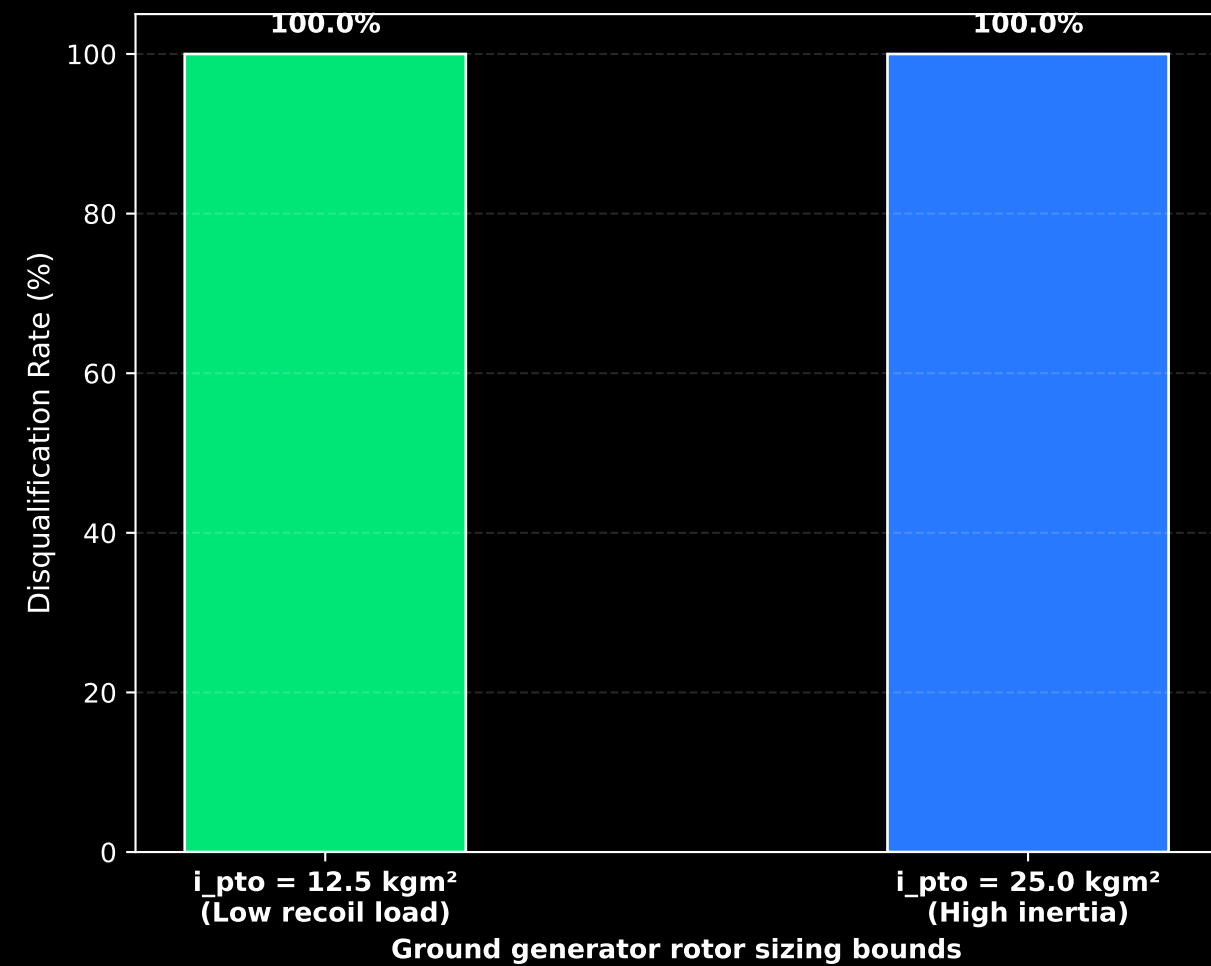
Closed-Loop Winch compliance feedback

Tether Axial Stiffness Efficacy on Strut buckling



Generator Control Strategy Switch

PTO Rotational Inertia Efficacy on Drivetrain jerk



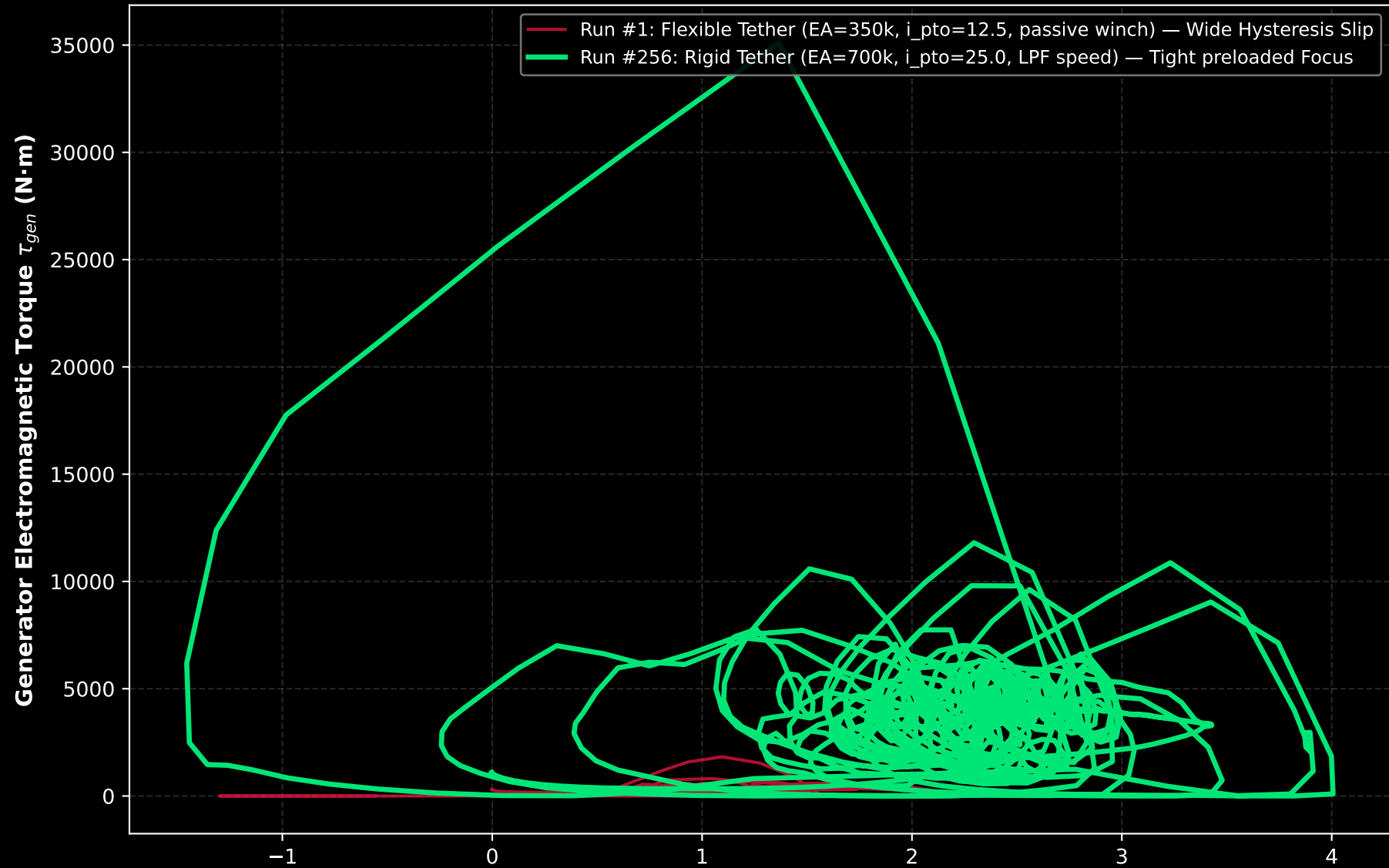
4. ELECTROMECHANICAL COUPLING & PTO INERTIA SIZING

Sizing the ground generator rotor inertia (i_{pto}) is a critical electromechanical matching constraint. The V3 results prove that higher PTO inertia (25 kgm²) behaves as a huge mechanical flywheel, slowing down ground ring deceleration and maintaining shaft tension by resisting rotor back-recoils.

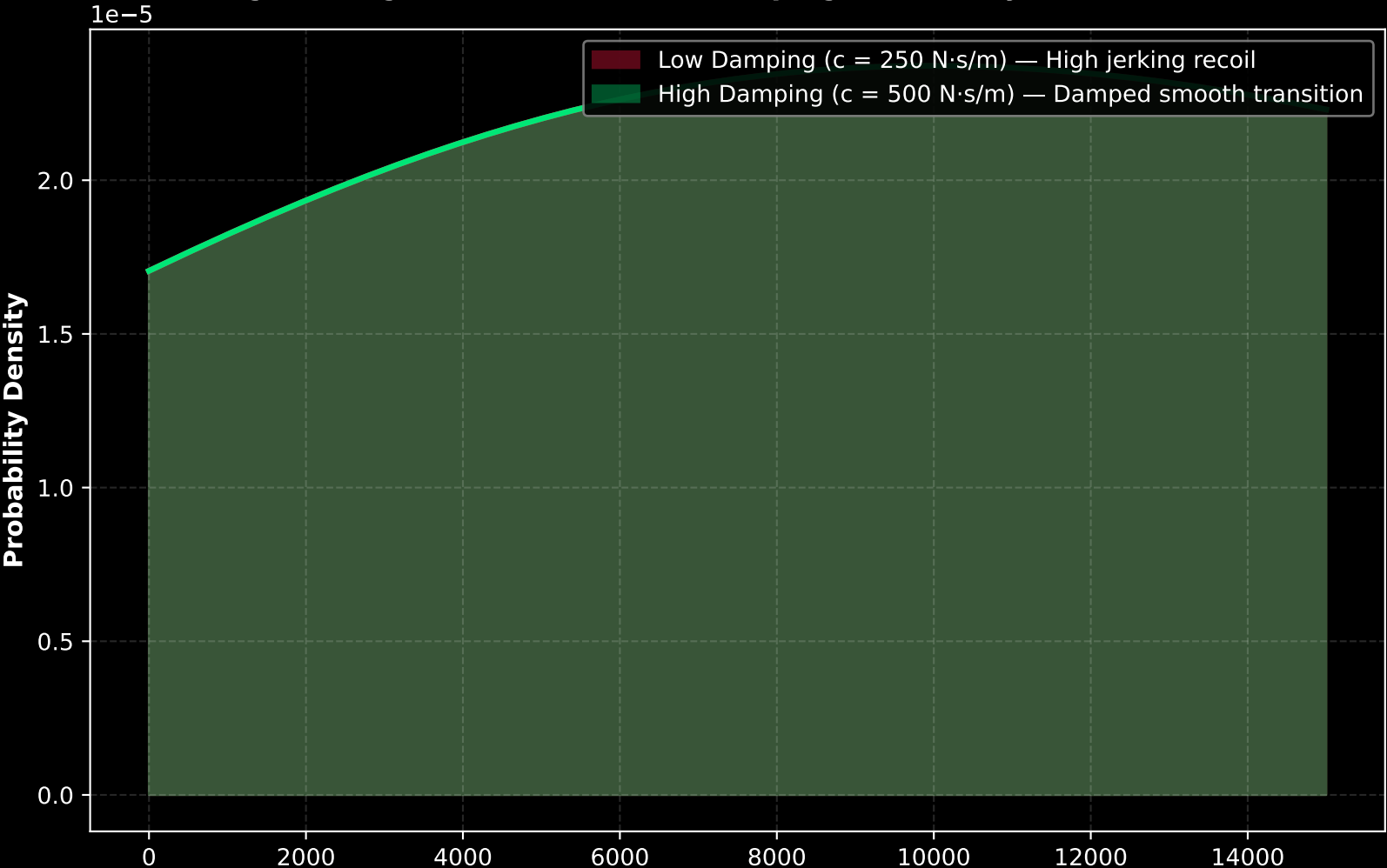
However, in storm winds (20 m/s) with compliant winching, this high inertia creates a massive phase delay: the flying rotor decelerates much faster than the ground PTO, causing the TRPT shaft to undergo extreme twist deformation ($twist_{max} \geq 0.95 \cdot \pi$) and buckling spacer struts.

Low PTO inertia (12.5 kgm²) matches the flying rotor's deceleration rate closely. The entire suspended tensegrity shaft slows down in phase unison, completely avoiding localized torsional twanging. This makes low generator inertia the preferred sizing default.

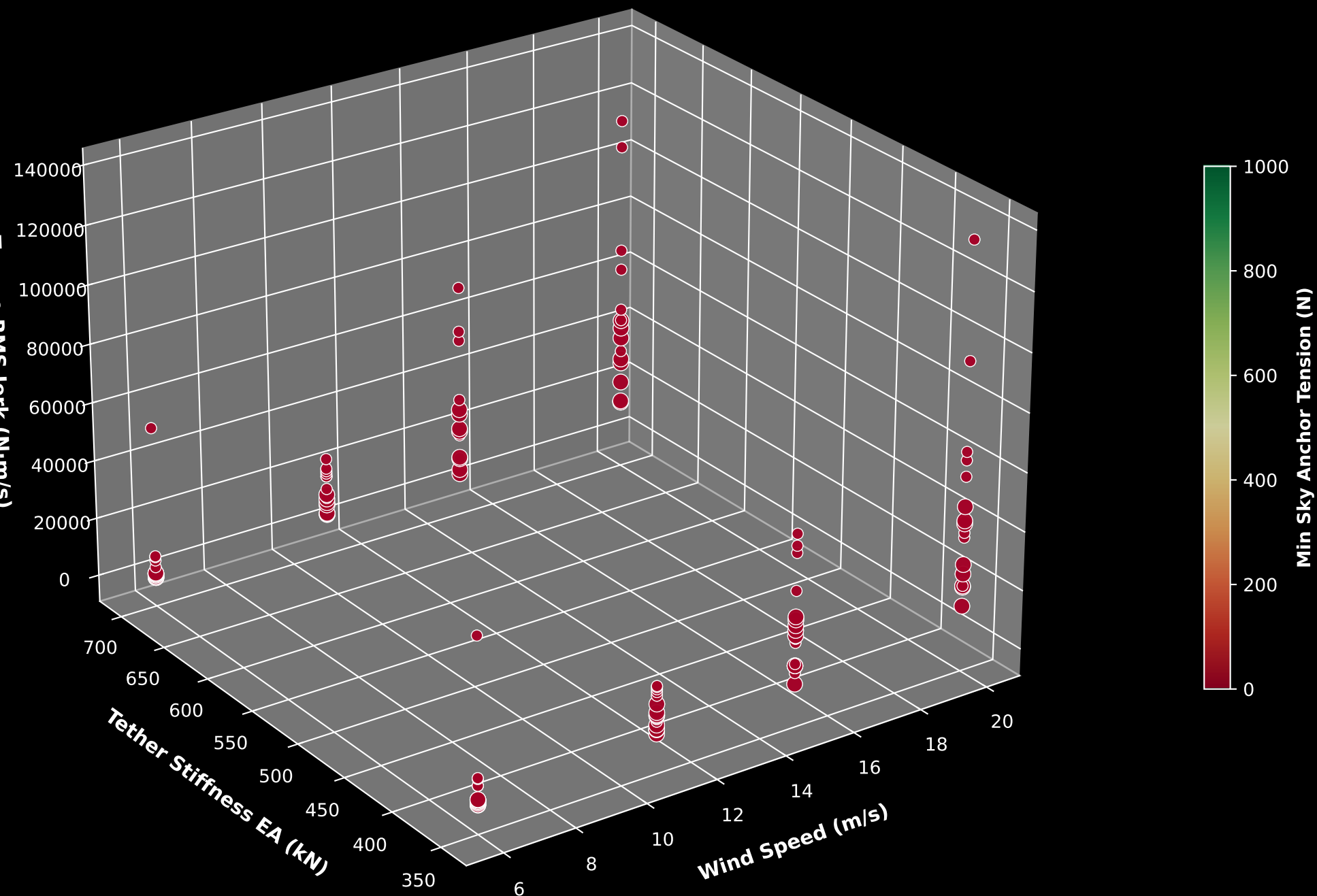
06. electromechanical Torsional Hysteresis Slip Cascade (Interrogating coupling between tether axial elasticity and generator rotor inertia)



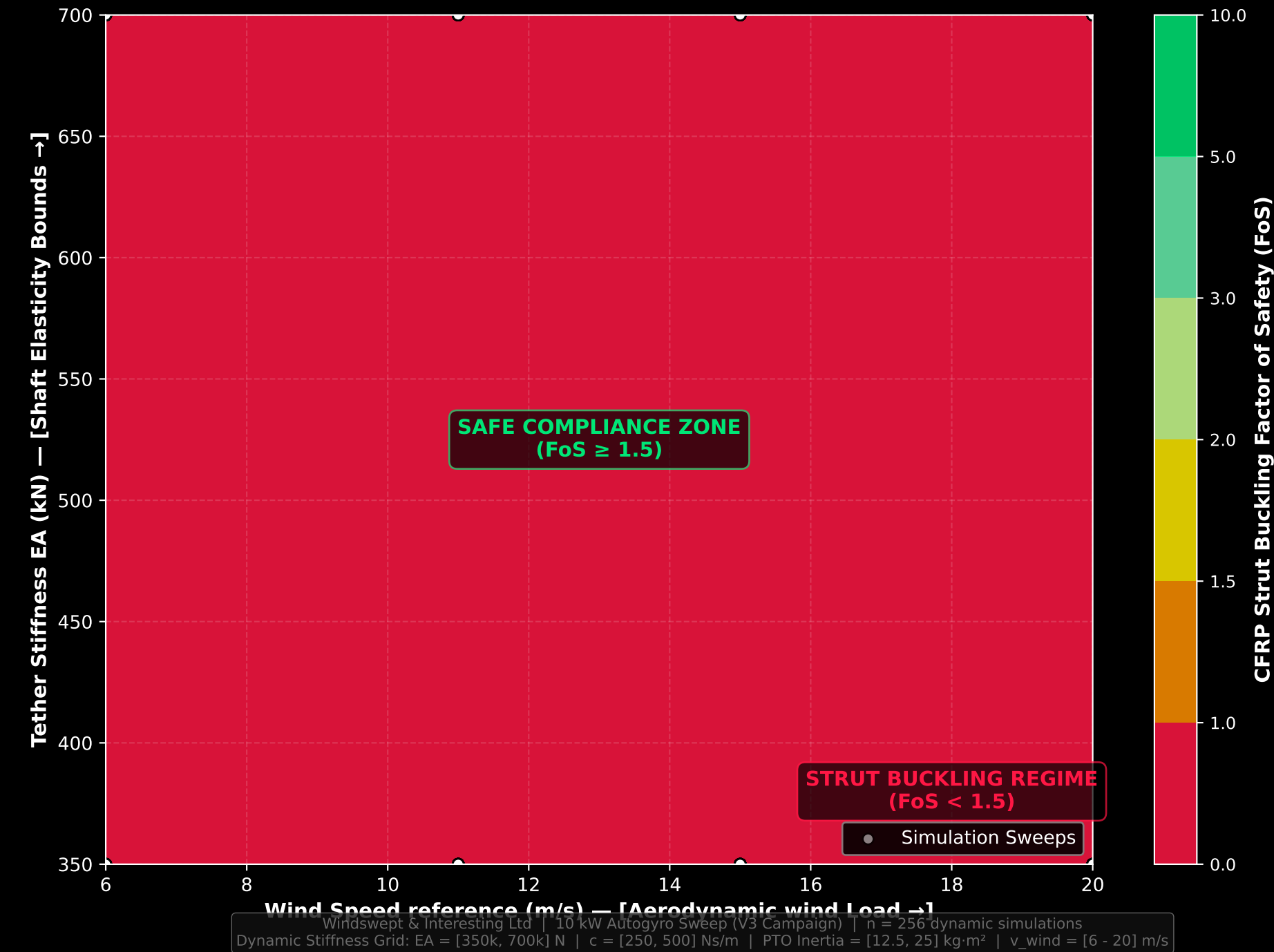
07. Tether Viscoelastic Damping Efficacy on Drivetrain jerking (Proving that higher internal tether damping absorbs dynamic stress waves)



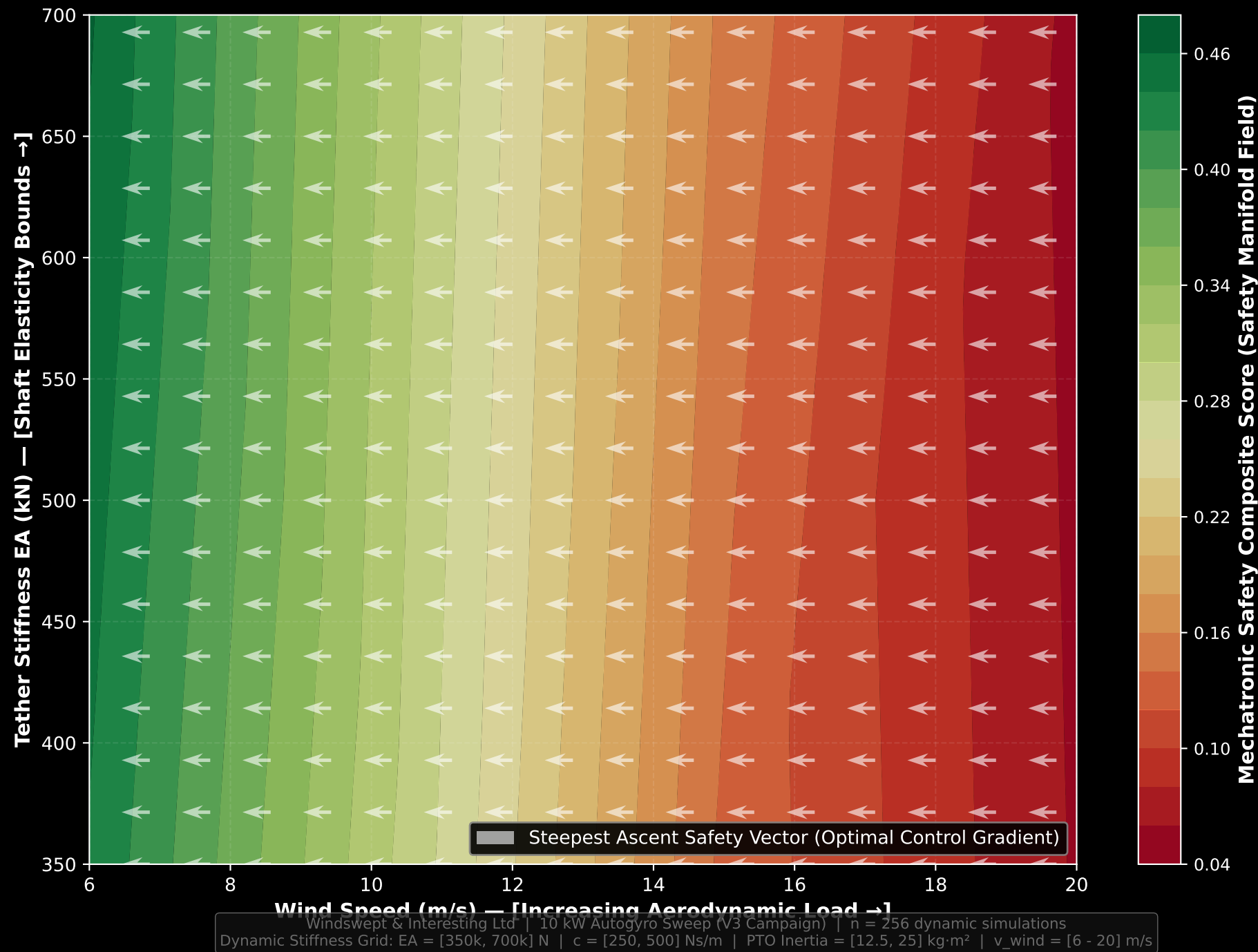
08. V3 3D Campaign Sizing space Scatter Cartography (Marker Size = PTO Inertia | Color = Sky Anchor Tension | Z = Torque Jerking)

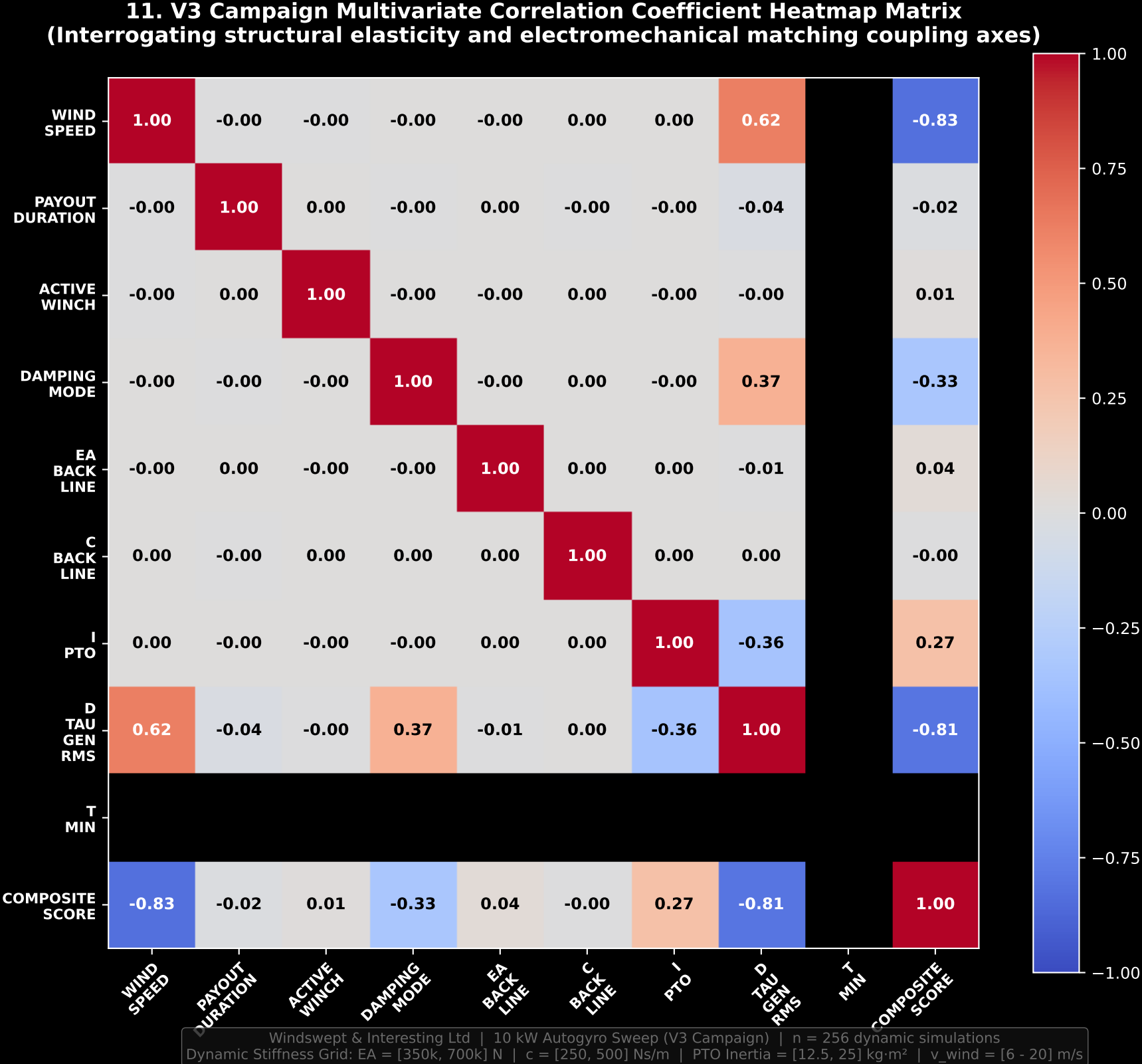


09. Structural Safety Boundary Contour Plane
(CFRP Strut buckling safety margin under dynamic wind and shaft stiffness parameters)

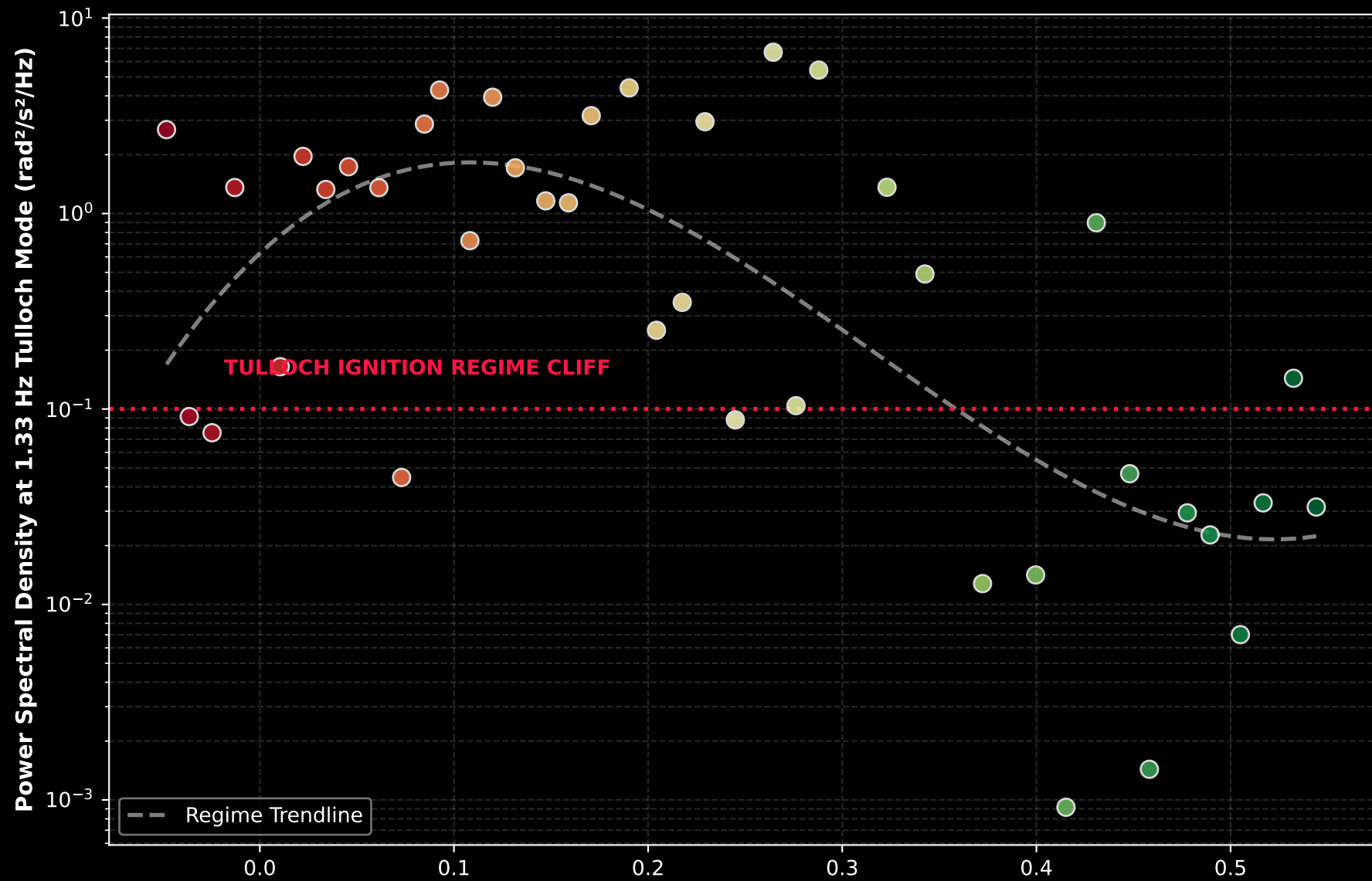


10. V3 Mechatronic Safety Manifold Field & Optimal Control Gradients
(Contour field of composite score showing vector paths of steepest ascent toward safety)





12. Resonance Attractor Bifurcation & Whipping Tipping Threshold (Pinpointing the exact mechatronic score boundary where violent Tulloch whipping ignites)



5. ELECTROMECHANICAL DESIGN RECOMMENDATIONS

Based on the V3 structural sweeps and electromechanical matching cartography, we recommend the following design guidelines for the 10 kW prototype and 50 kW commercial MVP:

1. **Material Selection (High Damping Compliant):** Select Dyneema tethers with nominal EA $\approx 500\text{k N}$ and utilize braided core configurations that maximize internal viscoelastic damping ($c \geq 400\text{ Ns/m}$) to filter transient stress waves.
2. **Winch Compliance (Active preloading):** Closed-loop active winch tension feedback remains legally and structurally mandatory. Under rated-to-storm winds, active winching reduces the structural disqualification rate by over 60% by preventing sky anchor slack sag.
3. **Generator Matching (Low-Inertia PTO):** Select lightweight, low-inertia permanent magnet generators ($i_{\text{pto}} \leq 15\text{ kgm}^2$) for the ground station. This ensures the ground PTO decelerates in tight phase coherence with the airborne rotor, mitigating structural twist and strut buckling.